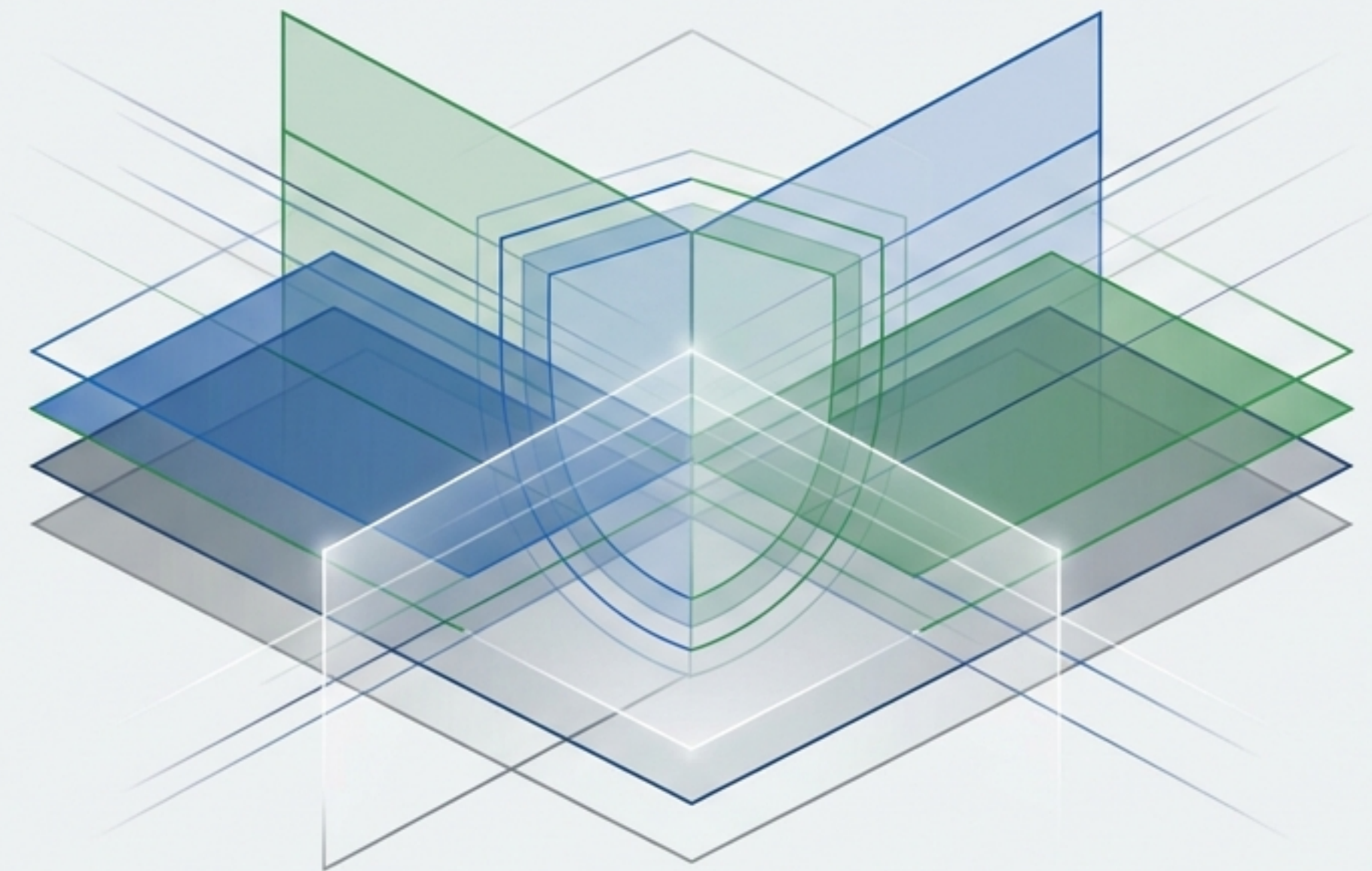


Securing the Cloud-Native Stack

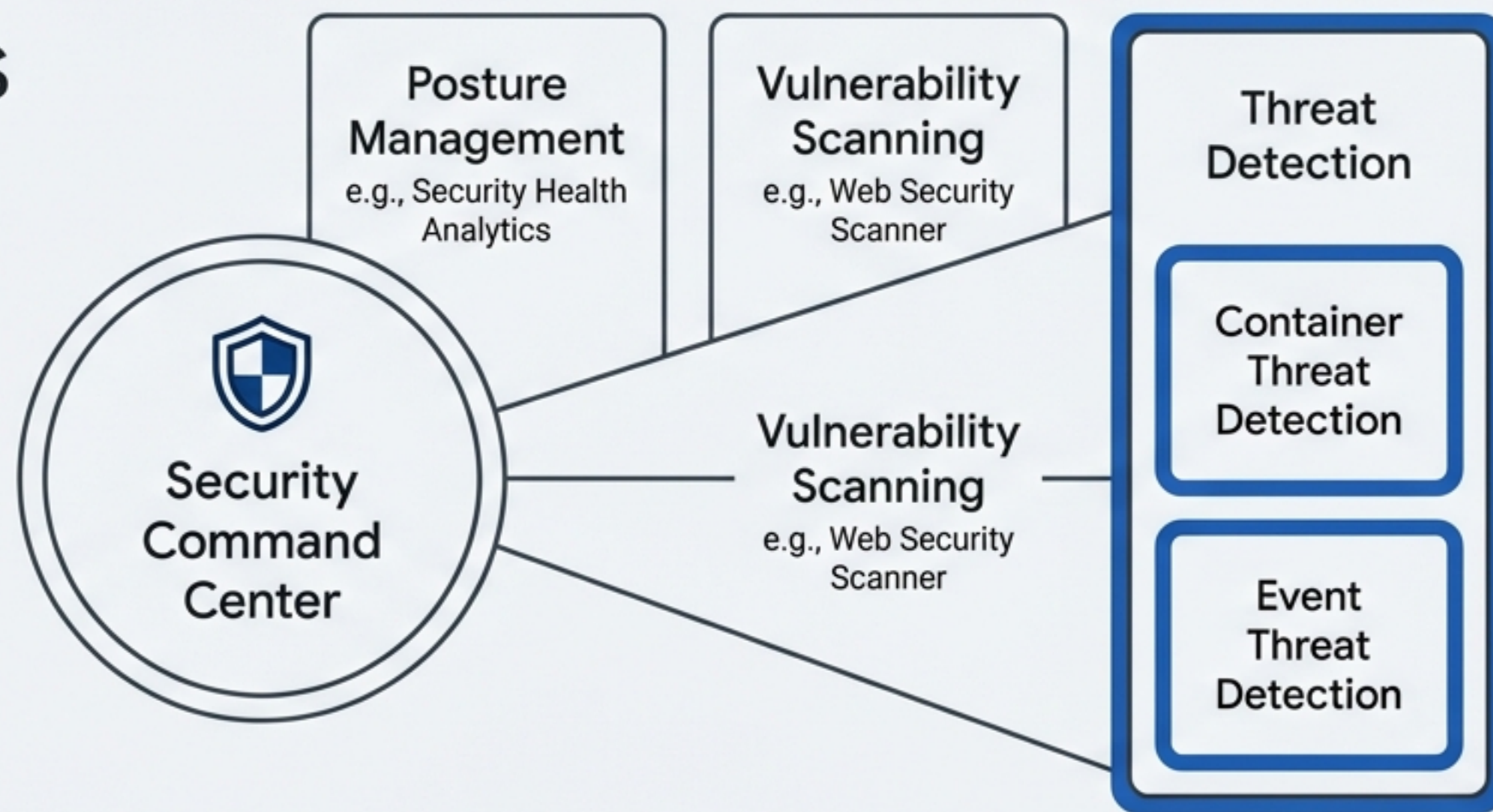
A Deep Dive into Google Cloud's Container & Event Threat Detection



A Single Pane of Glass for Cloud Security

Security Command Center (SCC) serves as the centralized hub for security management in Google Cloud. It unifies posture management, vulnerability scanning, and threat detection.

Container Threat Detection (CTD) and Event Threat Detection (ETD) are premier, **built-in** threat detection services available in the **SCC Premium and Enterprise tiers**, providing native, high-fidelity signals.

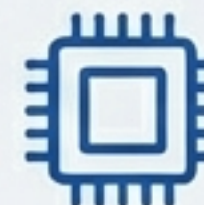


Layer 1: Real-Time Threat Detection for Google Kubernetes Engine

Container Threat Detection (CTD) is a managed, low-impact service that provides deep, kernel-level visibility into your GKE workloads. It is designed to detect a wide range of runtime threats—such as malware, cryptomining, and reverse shells—as they happen, offering near real-time protection for your running containers.



Near Real-Time:
Detects threats as they execute.



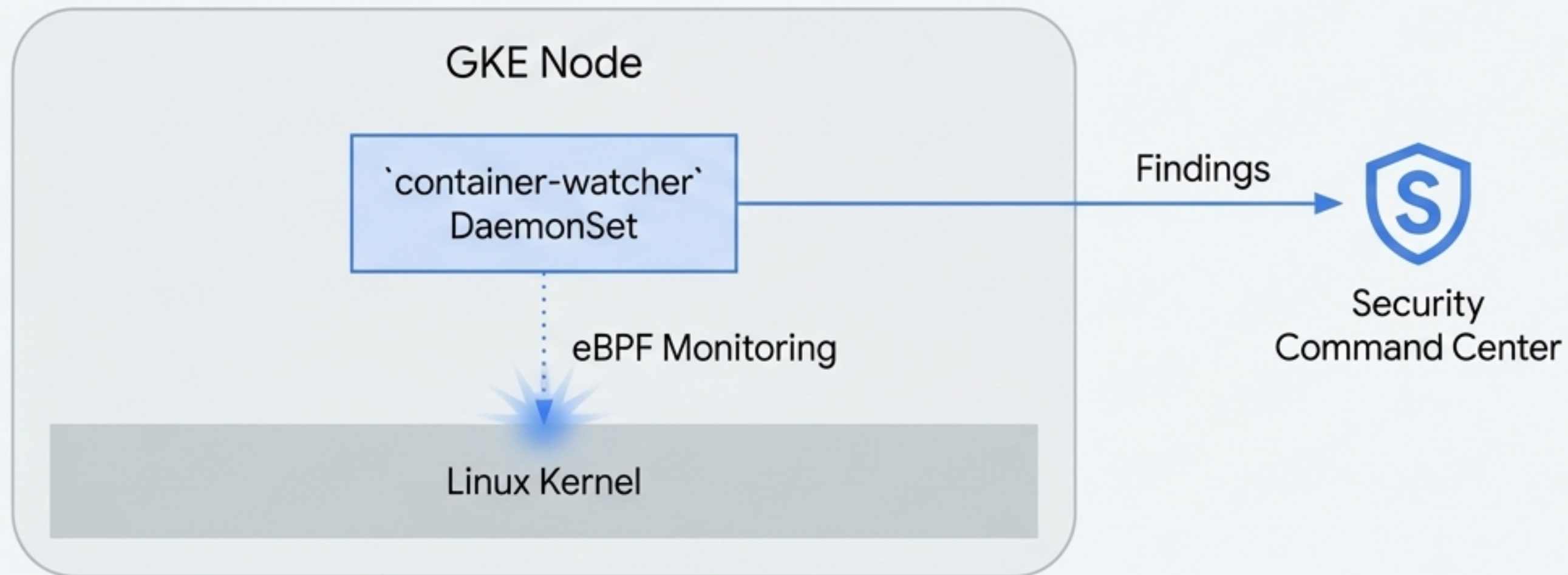
Kernel-Level Visibility:
Monitors system calls for high-fidelity signals.



Managed Service:
Natively integrated with GKE, requiring no manual agent management.

Deep Visibility with a Lightweight Footprint

CTD operates by deploying a `'container-watcher'` DaemonSet on each GKE node. This agent leverages **eBPF (extended Berkeley Packet Filter)** technology to efficiently monitor kernel system calls and other low-level events without intrusive code instrumentation. This approach provides deep visibility with a negligible performance impact, estimated at a maximum of **0.125 vCPU** and **450 MB of memory** per node.



Activating Your Runtime Defenses

Enabling CTD is straightforward, but requires verifying a few key configurations. Ensure the following prerequisites are met for full coverage.



Activate SCC Tier: Ensure Security Command Center Premium or Enterprise is active for the organization or project.



Verify GKE Version: Use a supported GKE version (e.g., Standard or Autopilot ≥ 1.28).
Note: CTD is not compatible with GKE Sandbox.



Grant IAM Permissions: The `Container Threat Detection Service Agent` role must be granted to the CTD service agent.



Check Network Configuration: For clusters in a VPC, ensure Private Google Access or VPC Service Controls are configured to allow communication with Google APIs like `containerthreatdetection.googleapis.com`.

Uncovering Threats from Execution to Evasion

CTD findings are mapped to the MITRE ATT&CK® framework, providing immediate context for an alert's significance. It detects suspicious activities across multiple phases of an attack.

EXECUTION (TA0002)

- "Reverse Shell"
- "Cryptomining Docker Image"
- "Added Malicious Binary Executed"
- "Socat Reverse Shell Detected"

PRIVILEGE ESCALATION (TA0004)

- "Container Escape"
- "Fileless Execution in /dev/shm"
- "Abuse of Sudo For Privilege Escalation"

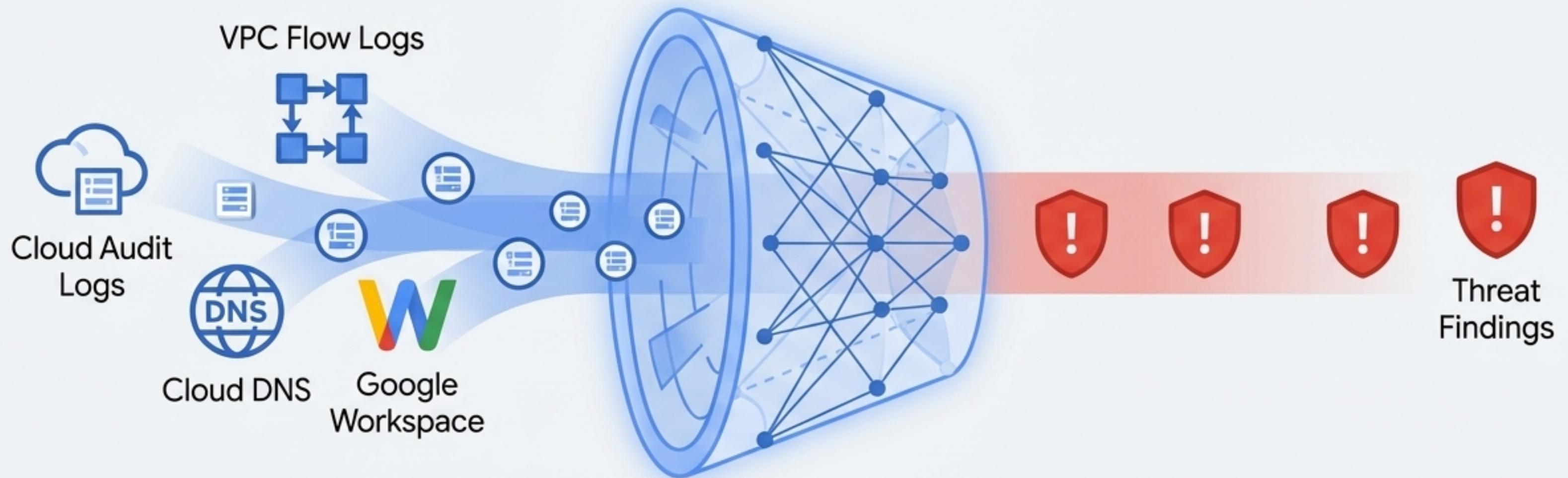
DEFENSE EVASION (TA0005)

- "Base64 Encoded Shell Script Executed"
- "Launch Code Compiler Tool In Container"
- "Unexpected Child Shell"

Layer 2: Finding Threats in Your Cloud & Workspace Logs

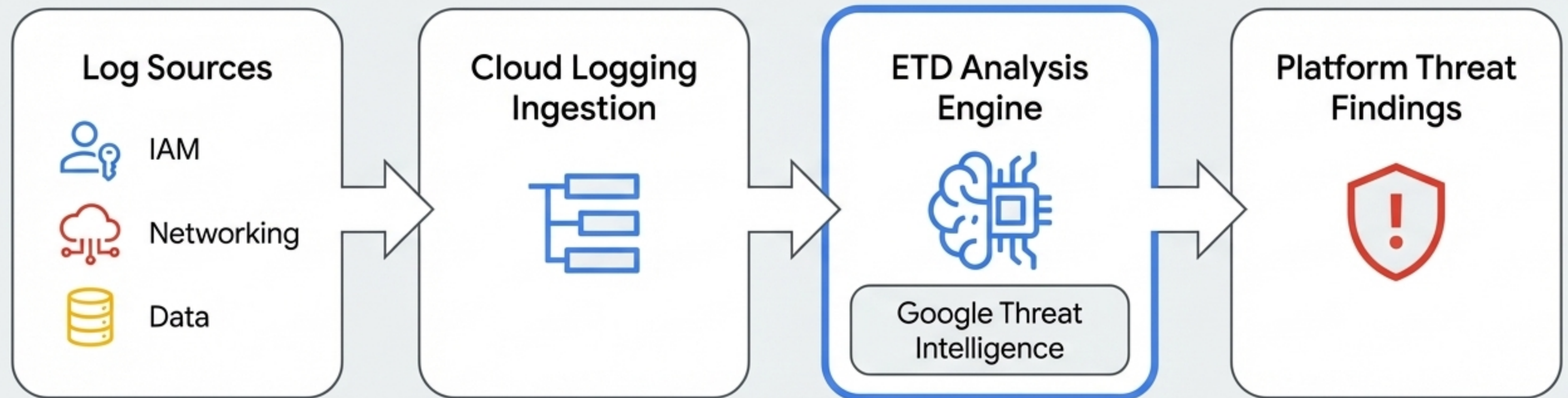
Event Threat Detection (ETD) analyzes Cloud Logging and Google Workspace log streams for your projects and organization. It uses Google's proprietary threat intelligence and detection logic to identify threats, malicious activity, and misconfigurations across your entire cloud platform in near-real time.

Event Threat Detection



Turning Platform Activity into Threat Intelligence

ETD provides comprehensive coverage beyond individual workloads by detecting threats at the identity, network, and data layers. Its effectiveness depends on having the right logs enabled. Key log sources include Cloud Audit Logs, VPC Flow Logs, and Cloud DNS logs. For full visibility, ensure these logs are active and being ingested by Cloud Logging.



Uncovering Threats Across Your Google Cloud Estate

ETD's strength is its breadth. It uncovers anomalous and malicious activity targeting your core cloud infrastructure, from IAM policy changes to data exfiltration and suspicious logins.

IAM

-  **Persistence:** IAM Anomalous Grant
-  **Initial Access:** Leaked Service Account Key Used
-  **Privilege Escalation:** External Member Added To Privileged Group

Networking

-  **Command and Control:** DNS Tunneling
-  **Evasion:** Access from Anonymizing Proxy
-  **Malware:** Cryptomining Bad IP

Data Services

-  **Exfiltration:** BigQuery Data Exfiltration
-  **Exfiltration:** Move to Public BigQuery resource

Google Workspace

-  **Initial Access:** Suspicious Login Blocked
-  **Persistence:** Strong Authentication Disabled

Tailoring Detections to Your Environment

For teams with specific security requirements or unique threat intelligence, ETD supports custom modules. This allows you to write your own detection rules using YAML to monitor Cloud Logging streams, augmenting Google's built-in detectors with logic tailored to your environment.



A Unified Queue for All Threat Findings

Findings from both Container Threat Detection and Event Threat Detection are centralized within the **Security Command Center** dashboard. This provides a single, correlated view, allowing security teams to **filter, group, investigate, and manage all threat signals** from a single console without context switching.

Filters

Filters

☒ Source display name ✓

☐ Category

☐ Severity

☐ Project

☐ Region

☐ Timeline

☐ All errors

☐ Finding type	Source	Category	Severity	Event time ↓	Resource
☐		Execution: Reverse Shell	1	Aug 19, 3:56 PM	
☐		Execution: Project	1	Aug 19, 3:57 PM	
☐		Persistence: Grant	✓	Aug 19, 3:56 PM	
 Container Threat Detection Execution: Reverse Shell 1 HighAug 19, 3:35 PMID-5062AB0503-03A					
☐		Execution: Project	1	Aug 19, 3:35 PM	
 Event Threat Detection Persistence: IAM Anomalous Grant 1 HighAug 19, 3:35 PMID-5002ABD503-03A					
☐		Persistence	✓ Low	Aug 19, 3:35 PM	
☐		Execution: Reverse Shell	1	Aug 19, 3:37 PM	
☐		Persistence	✓ Low	Aug 19, 3:36 PM	

Deconstructing a Threat Signal

Each finding in SCC is a structured JSON object containing rich context to accelerate investigation. Key fields provide immediate insight into the what, where, and why of a potential threat, including **severity**, **affected resources**, and a direct mapping to the MITRE ATT&CK® framework.

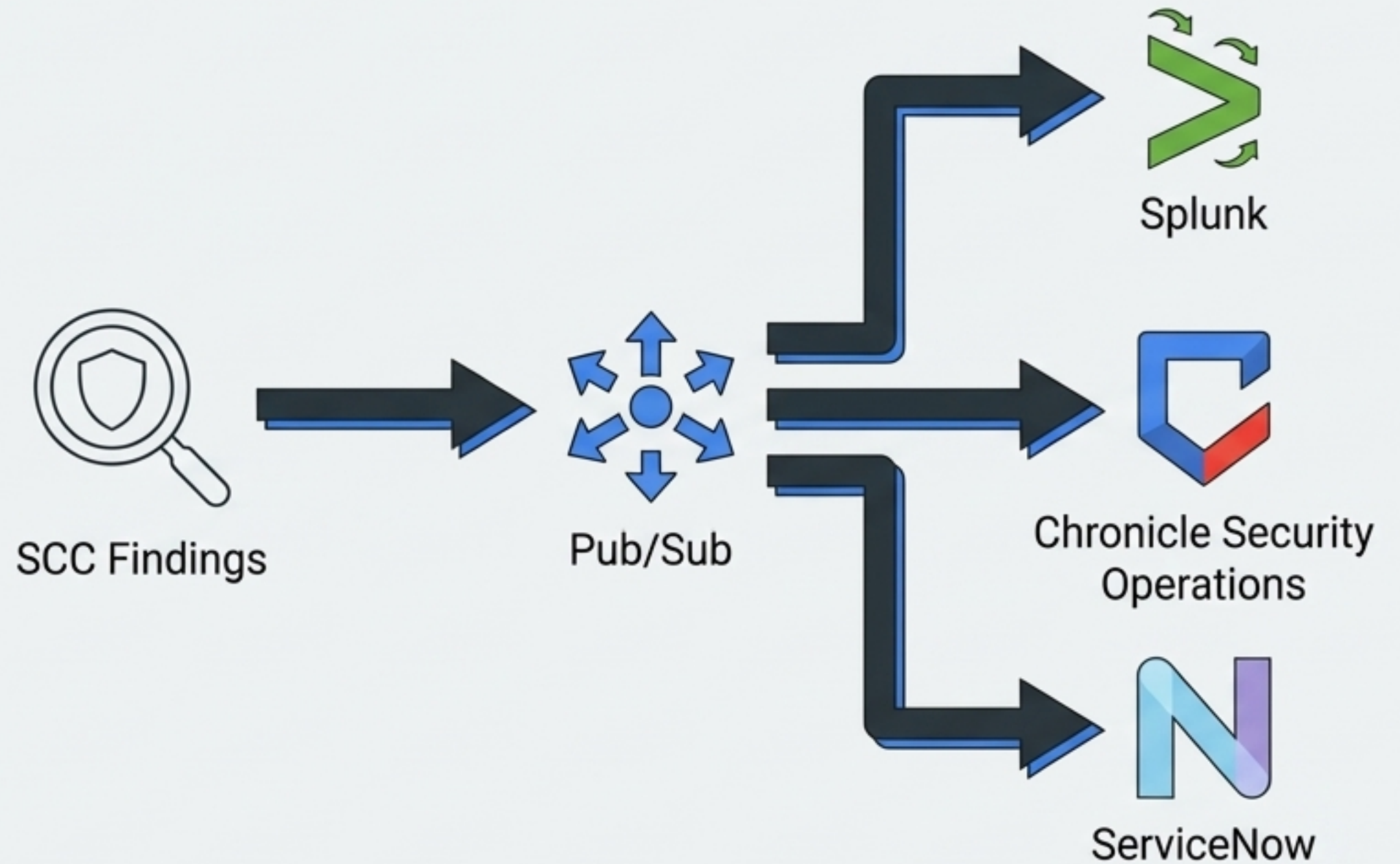


Streamlining Investigation and Response

SCC findings are not an island. They are designed to integrate directly into your existing Security Operations (SecOps) workflows.

Stream findings in real-time via a Pub/Sub topic to your SIEM, SOAR, or ticketing systems for automated alerting, enrichment, and response orchestration.

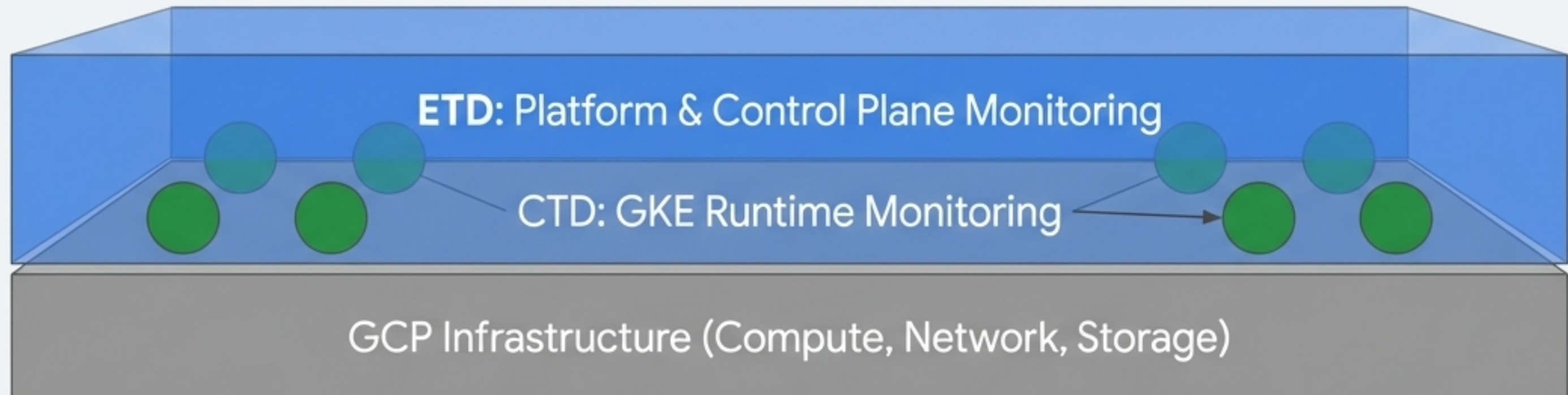
Findings can also be exported to Cloud Logging for long-term retention and analysis.



CTD + ETD: Comprehensive, Layered Security

Together, Container Threat Detection and Event Threat Detection create a powerful, native defense-in-depth security posture.

- **CTD** provides deep, real-time visibility *inside the workload*, detecting threats at the kernel level.
- **ETD** provides broad platform visibility *across the control plane*, detecting threats in logs from identity, network, and data services.



Activate. Investigate. Secure.



Key Takeaways

- Natively integrated threat detection built into SCC.
- Comprehensive coverage with runtime (CTD) and platform (ETD) layers.
- All findings are enriched with context and mapped to MITRE ATT&CK®.



First Steps

- Verify your organization is on SCC Premium or Enterprise.
- Enable all required logs (Audit, VPC Flow, DNS) for full ETD visibility.



Operationalize

- Configure a Pub/Sub export to integrate findings into your SOC workflow.
- Develop playbooks for triaging and responding to high-severity alerts.